

Master Control Lead Scoring and Conversion Optimization

Predicting high-value leads to boost sales efficiency



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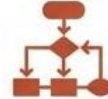
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Impact

ROI & action plan

Product Portfolio Performance Gap

Mx (Manufacturing)

12.7%

Conversion Rate

7% Gap:
Revenue Loss

\$

Qx (Quality)

19.7%

Conversion Rate

Sales efficiency impact



The Problem

SDRs operate with limited time and lack a data-driven system to prioritize which leads to contact first, resulting in inefficient resource allocation.



Cost Reality

Each sales call costs \$50 in SDR time and resources, while each successful conversion generates \$6,000 in value.



Revenue Impact

The 7% conversion gap between Mx and Qx translates directly to substantial lost revenue and missed growth opportunities.



Strategic Need

Intelligent lead prioritization is essential to maximize ROI by directing SDR efforts toward highest-probability conversion opportunities.

Project Objective



Predictive Model

Build a machine learning lead scoring model to identify and rank high-probability conversion opportunities using historical CRM data.

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Data-Driven Ranking

Generate probability scores for all leads and rank them by conversion conversion likelihood, enabling systematic prioritization.



ROI Maximization

Optimize sales team efficiency and revenue impact by directing SDR outreach to the right leads at the right time.

Dataset Overview & Key Traits



Scale

Approximately 16,800 leads analyzed across Mx and Qx product lines



Imbalance

Severe class imbalance with ~85% non-converted leads requiring specialized modeling techniques



Conversion

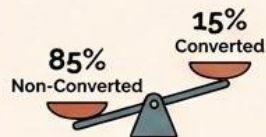
Overall conversion rate of 18% across the entire dataset



Performance Gap

Distinct 7% performance gap between Mx (12.7%) and Qx (19.7%) segments

Key Technical Challenges



Severe Class Imbalance

- **85% non-converted** leads required **SMOTE**, class weighting, and threshold optimization techniques.



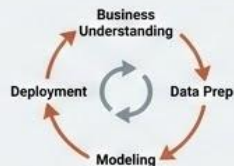
Categorical Feature Dominance

- Limited numerical data necessitated advanced **encoding strategies** and feature interaction engineering.



Business Explainability Requirement

- Sales adoption required **SHAP values** and interpretable **feature importance** for actionable insights.

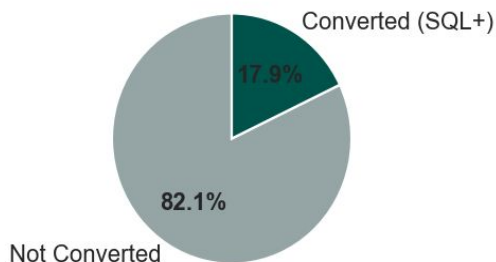


CRISP-DM Framework

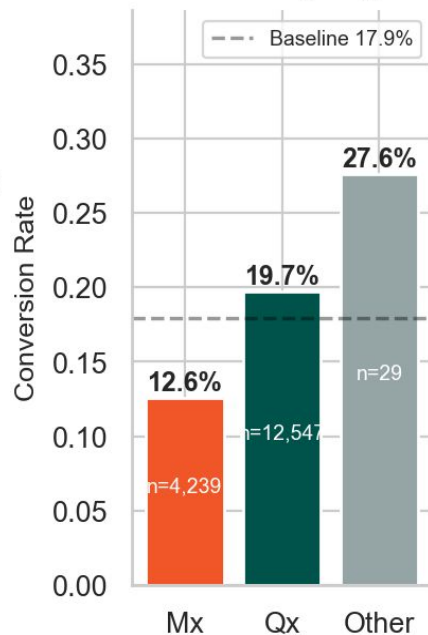
- Structured methodology ensured **business understanding**, data preparation, modeling, and deployment alignment.

Class Imbalance in the Data

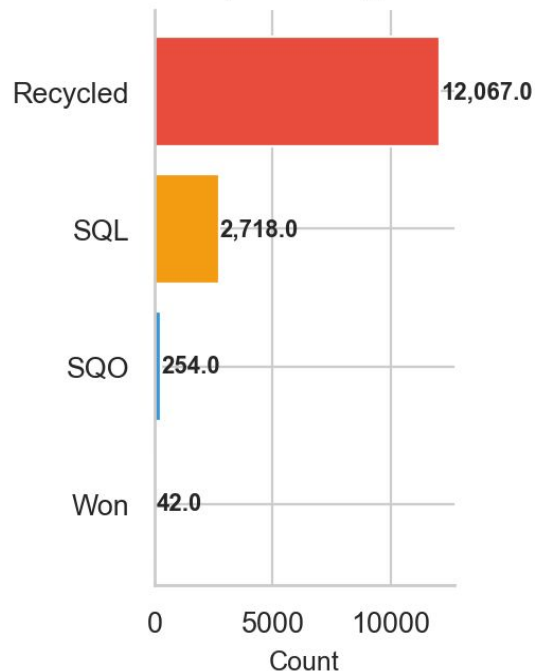
Target Distribution



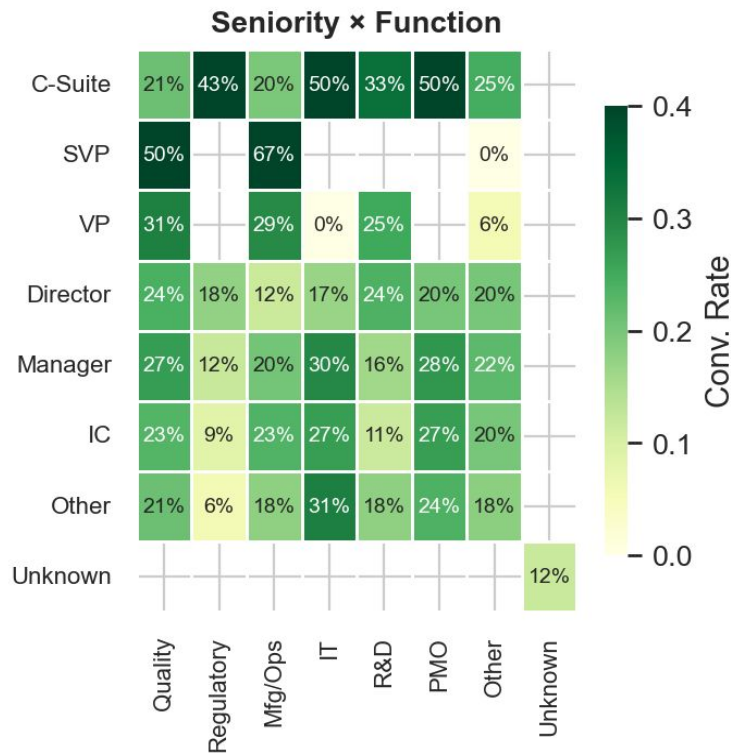
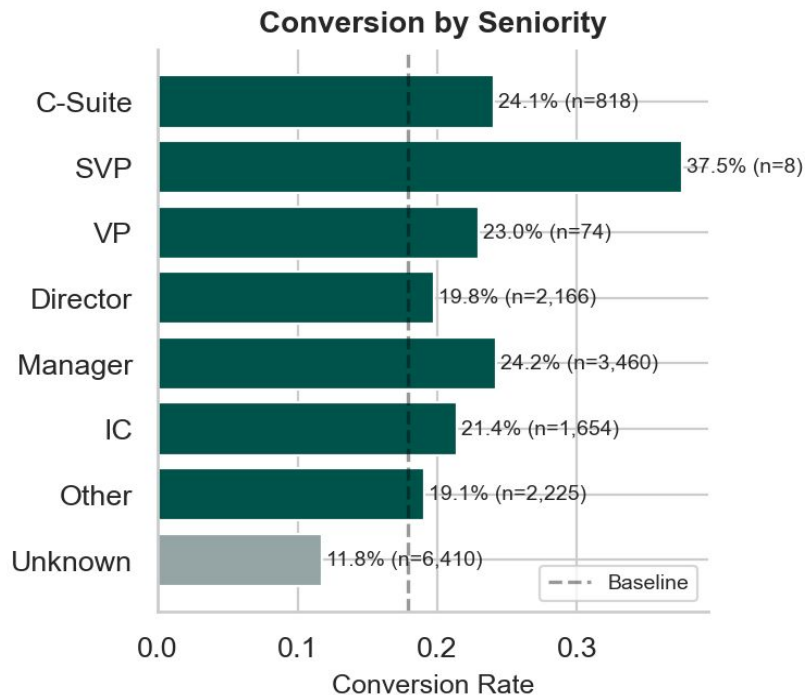
Conversion by Segment



Pipeline Stages



Free-Text Job Title Complexity



Core Feature Engineering Strategy

 Intent Strength	Content downloads, demo requests, webinar attendance signal buying readiness
 Channel Efficiency	Premium channels (webinars, direct outreach) vs. low-value channels (organic search)
 Firmographics	Company size and industry segmentation identify ideal customer profiles
 Role Seniority	Decision-making authority indicators (VP, Director, C-suite) predict conversion
 Behavioral Recency	Time-based engagement signals capture momentum and buying cycle timing

Key Outcome: Strong separation achieved between converted and non-converted lead cohorts

Advanced Feature Engineering

Buying Committee Activity

Multiple stakeholders from same organization signal organizational buying intent

Cross-Product Interest

$Mx + Qx$ combined engagement indicates broader solution needs and higher deal value

Executive Momentum

Seniority $\times$ intent interaction captures executive-level buying signals and urgency
Industry compliance pressure (Pharma ra, extract decision-making Medical Device) predict quality system needs

Regulatory Urgency Scoring

Job title embeddings and role classification extract decision-making authority from unstructured text



These advanced features significantly improved model discrimination power

Exploratory Data Analysis: Key Findings

Logistic Regression	Baseline model providing interpretability and coefficient-based feature importance
Random Forest	Non-linear pattern detection with feature interaction discovery
Gradient Boosting Suite	XGBoost, LightGBM for advanced tree-based sequential error correction
CatBoost	Optimized for categorical features with built-in encoding strategies
Ensemble Techniques	Stacking and voting methods combined model strengths for optimal performance

Modeling Approach & Selection



Logistic Regression

Baseline model providing interpretability and coefficient-based feature importance



Random Forest

Non-linear pattern detection with feature interaction discovery



Gradient Boosting Suite

XGBoost, LightGBM for advanced tree-based sequential error correction



CatBoost

Optimized for categorical features with built-in encoding strategies



Ensemble Techniques

Stacking and voting methods combined model strengths for optimal performance

Model Evaluation Strategy



Primary Metric

AUC-ROC measures ranking performance and lead prioritization effectiveness



Metrics

Precision, Recall, F1-score, and calibration analysis ensure balanced performance



Generalize

Rigorous test set validation prevents overfitting and ensures real-world applicability



Business Fit

Prioritized actionable insights over raw accuracy for sales team adoption

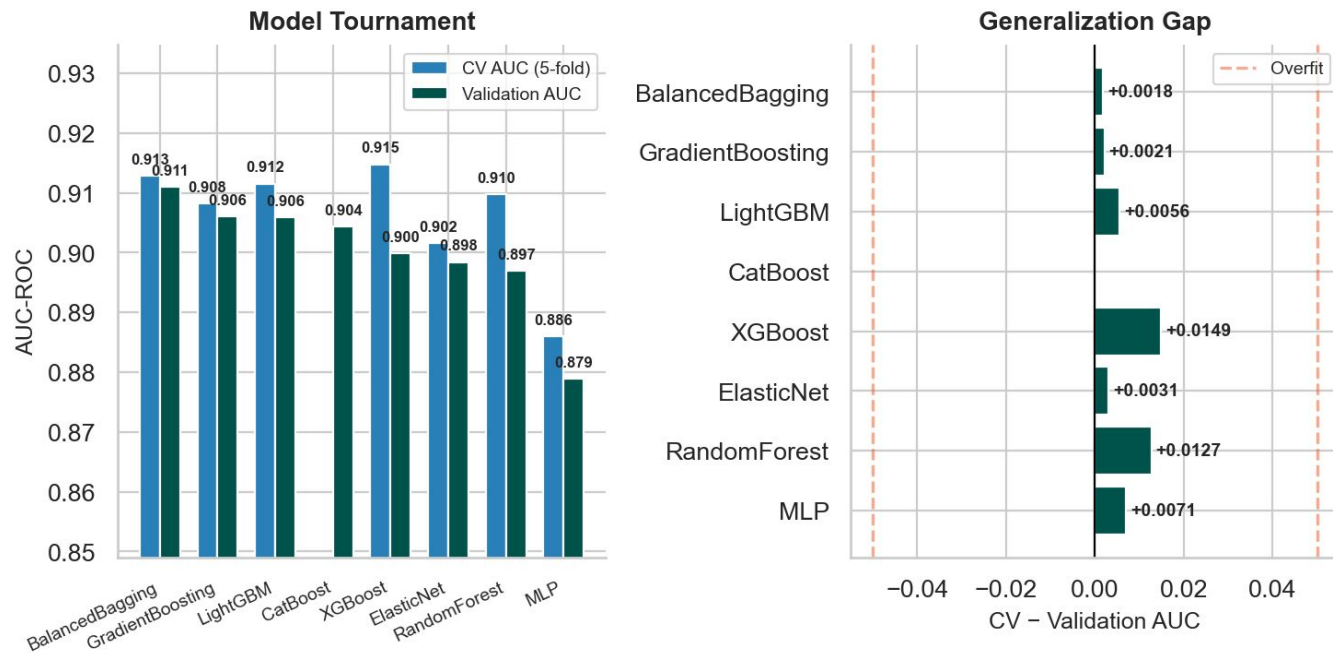


Cross-Validation

Stratified k-fold validation ensures robust performance estimation across data splits

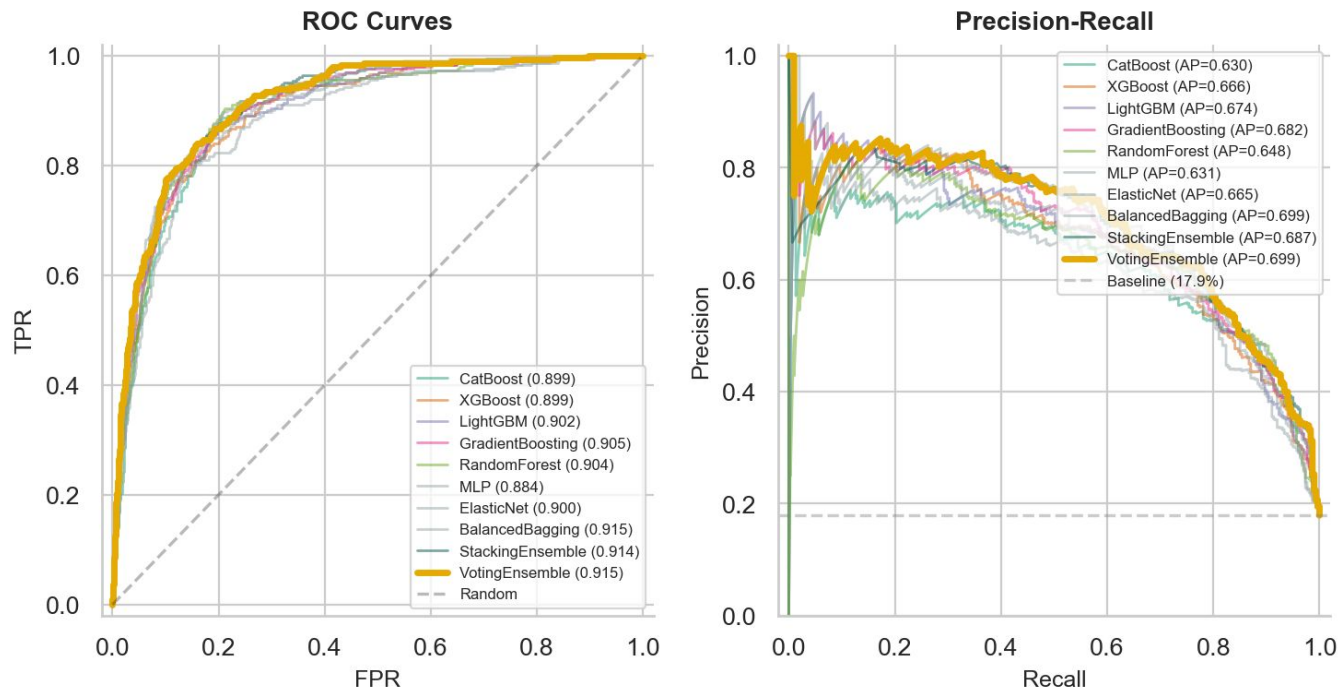
Model Evaluation Strategy Visualization

Part 1



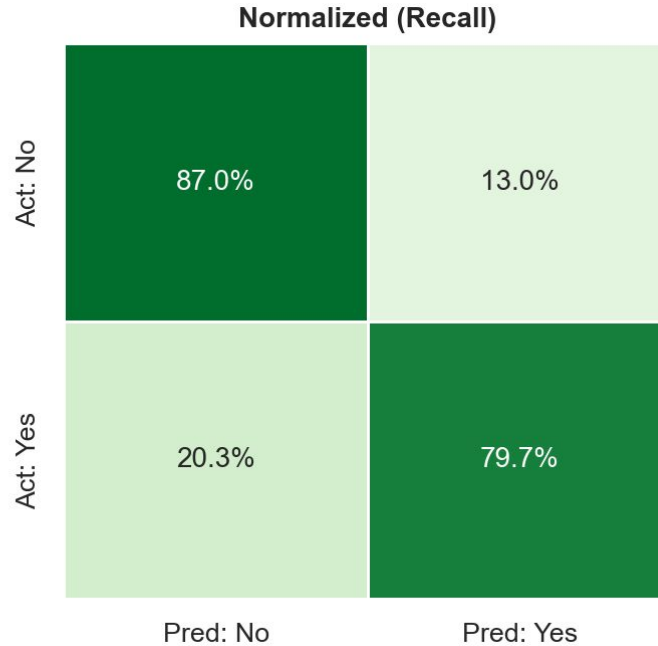
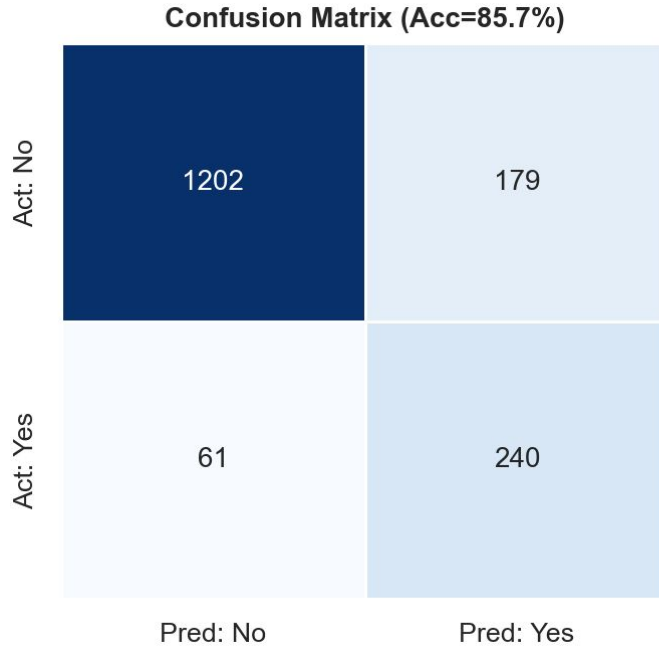
Model Evaluation Strategy Visualization

Part 2



Model Evaluation Strategy Visualization

Part 3



Model Explainability: SHAP Analysis



Senior Decision-Makers

VP, Director, and C-suite roles are the strongest conversion drivers

- ✓ Incremental conversion gains
- ✓ Untapped opportunities
- ✓ Cost-effective wins



Quality & Regulatory Roles

Highest conversion probability among all functional areas

- ✓ Regulatory compliance need
- ✓ Quality systems; high conversion potential



Premium Channels

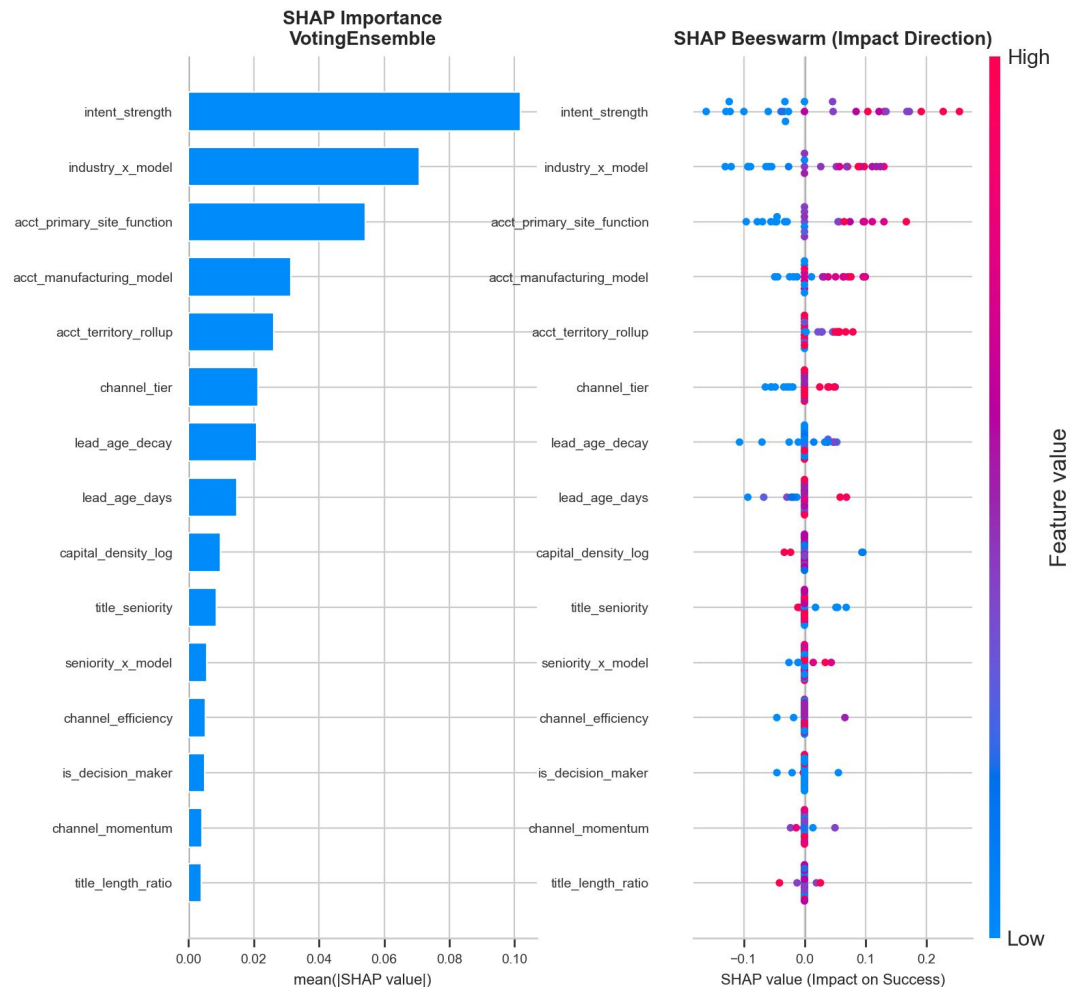
Webinars, demos, and direct outreach significantly outperform low-value channels

- ✓ Premium webinars
- ✓ Personalized demos
- ✓ Direct executive outreach

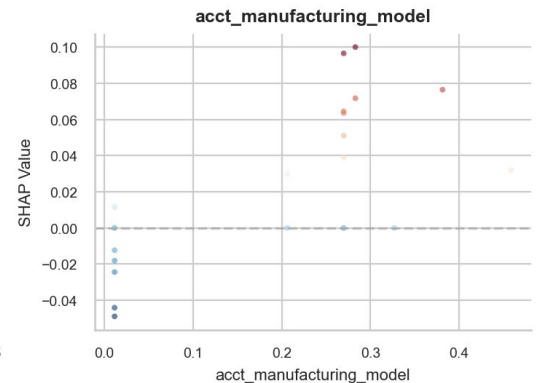
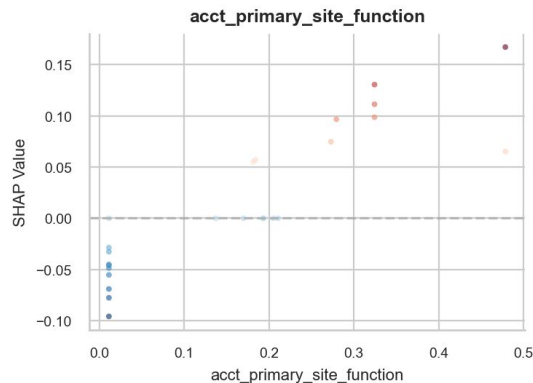
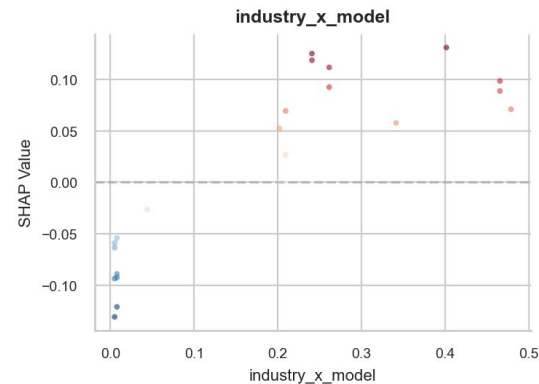
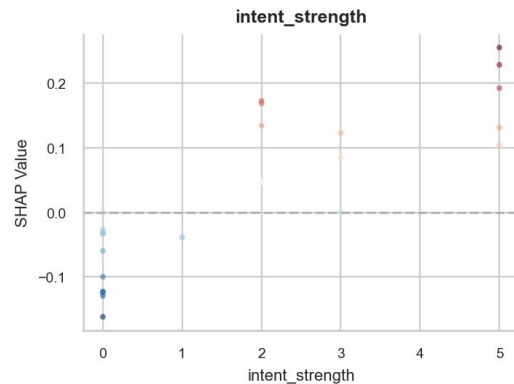
SHAP values reveal which features drive predictions, ensuring transparency and trust for sales adoption

SHAP Analysis

Visualization Part 1



SHAP Analysis Visualization Part 2



Business Impact Quantification



Scoring

Model generates probability scores for all leads, ranking conversion likelihood.

Strategy

Ranked lead lists enable optimal SDR contact strategy and time allocation.

Selection

Profit maximization through data-driven lead selection and prioritization.

Allocation

Resource allocation optimized based on conversion likelihood scores.

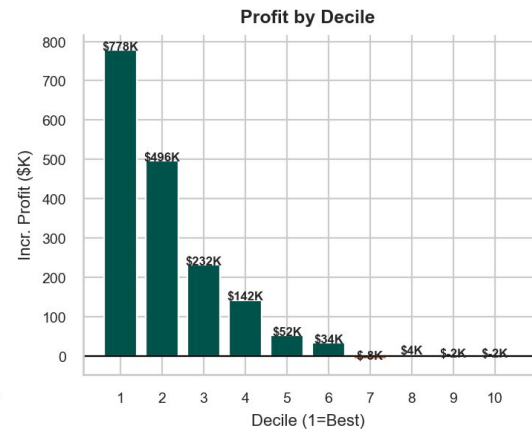
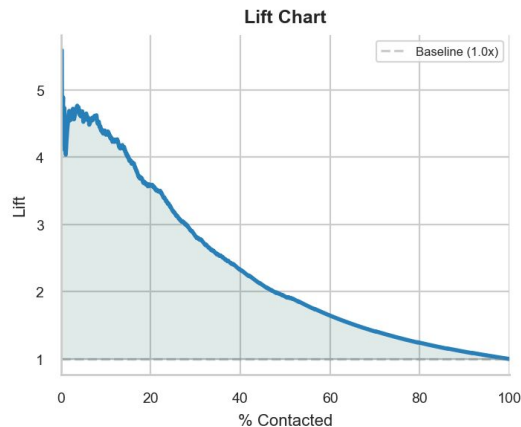
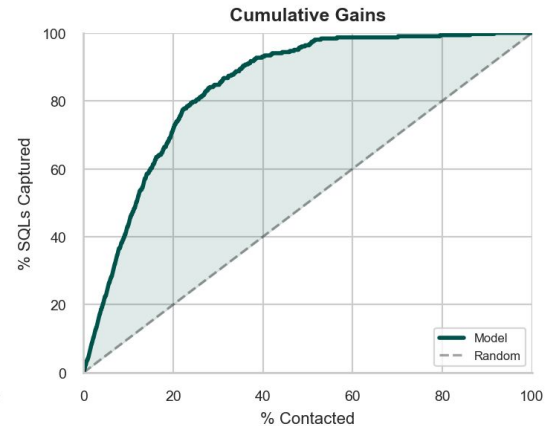
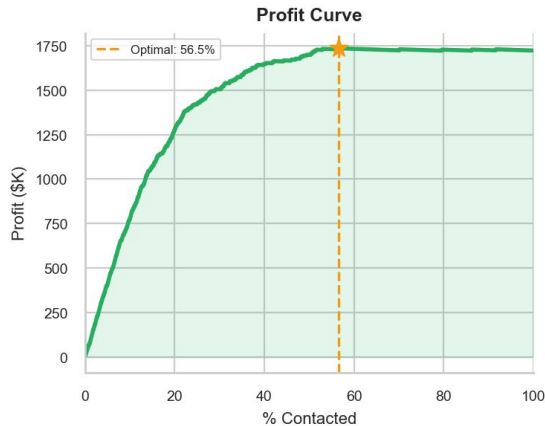
ROI

Measurable ROI improvement through targeted outreach to high-probability leads.

Business Impact Visualization

Optimal Profit Configuration

Metric	Value
Optimal Threshold	0.076
Leads to Contact	951
SQLs Captured	297
% Population	56.5%
SQL Capture Rate	98.7%
Net Profit	\$1,734,450
Lift vs Random	1.7x



Final Model Summary KPIs

Metric	Value
Test AUC-ROC	0.9147
Average Precision	0.6993
Best Model	VotingEnsemble
Optimal Threshold	0.076
Max Profit (Test)	\$1,734,450
Annual Lift	\$150,600
Calls Required	951 (57%)
SQL Capture	99%
Top Decile Conv.	77.5%
Baseline Conv.	17.9%

Strategic Recommendations

Priority

Focus SDR time exclusively on top-ranked leads



- High-probability leads
- Maximize conversion, cut wasted outreach

Industry

Target Pharma and Biotech sectors



- Regulatory compliance need
- Quality systems; high conversion potential

Seniority

Engage senior decision-makers only



- VP and Director levels
- C-suite contacts
- Budget authority holders

Channels

Invest in high-intent engagement channels



- Premium webinars
- Personalized demos
- Direct executive outreach

Re-engagement

Re-engage high-scoring recycled leads



- Incremental conversion gains
- Untapped opportunities
- Cost-effective wins

Conclusion and Key Takeaway



Actionable Sales Intelligence

Data-driven lead scoring transforms CRM data into actionable intelligence for sales teams



Closes the Conversion Gap

Improves Mx targeting precision and closes the 7% conversion gap with Qx



Direct Revenue Impact

Delivers measurable revenue impact by connecting SDRs with the right leads at the right time



Scalable Framework

Provides a scalable framework for continuous optimization and learning as new data emerges

